E-COMMERCE SECURITY PROJECT

E-Commerce Web Application Development & Penetration Testing

6-WEEK PROJECT ROADMAP

Build → Test → Report → Remediate → Improve & Verify

Project Type	Security Focus
E-Commerce Web Application	Web Application Penetration Testing
Duration 6 Weeks	**Final Output** Tested, Remediated & Improved Web Application

ACADEMIC PROJECT • WEB DEVELOPMENT + CYBERSECURITY

1. Executive Summary

The project focuses on developing a complete E-Commerce Web Application and then performing a structured penetration test against the application. The team will first build the website and its core functionality, improve its design, perform security testing, report the identified bugs and vulnerabilities, and finally apply remediation and verify the fixes.

Project objective: deliver an e-commerce application that is functional, visually improved, tested against common web application security issues, and remediated based on the findings.

2. Scope & Rules of Engagement

• Testing is performed only against the team's own e-commerce application and authorized test environment.
• Testing activities are limited to the project scope and are intended for academic / lab use.
• Vulnerabilities are documented with evidence, impact, and recommended remediation.
• After fixes are applied, the relevant tests are repeated to verify the remediation.

3. Technologies & Tools

Category	Examples / Purpose
Frontend	HTML, CSS, JavaScript — interface and user experience
Backend	PHP / server-side framework — application logic and APIs
Database	MySQL / phpMyAdmin — products, users, orders and application data
Testing	Burp Suite, browser developer tools, manual security testing
Discovery	Nmap and other authorized enumeration / reconnaissance tools
Documentation	Screenshots, bug/vulnerability reports, remediation notes and final documentation

4. Six-Week Timeline

The project follows a practical sequence: build the application, improve its design, test security, report findings, then remediate and verify.

Week 1 | 10/15 | Website Building — Foundation

Set up project structure, database, authentication, product structure and core pages.

Week 2 | 10/22 | Website Building — Features

Complete shopping flow, product pages, cart, checkout, user features and backend functionality.

Week 3 | 10/29 | Website Design Improvement

Improve layout, responsiveness, usability, visual consistency and the overall e-commerce experience.

4. Six-Week Timeline — Continued

Week 4 | 11/5 | Penetration Testing

Perform reconnaissance and manual web security testing; identify vulnerabilities and collect evidence.

Week 5 | 11/12 | Reporting Bugs & Vulnerabilities

Document confirmed bugs and security findings with evidence, impact, severity and recommended fixes.

Week 6 | 11/19 | Remediation

Fix confirmed security vulnerabilities and functional bugs, apply secure coding / configuration changes, and retest the relevant issues to verify the fixes.

5. Team Roles

Member	Primary Role	Main Responsibilities
Member 1	Web Development Lead	Project structure, frontend pages, core e-commerce functionality and integration.
Member 2	Backend & Database	Server-side logic, database design, authentication, products, orders and data handling.
Member 3	Penetration Testing	Reconnaissance, vulnerability discovery, request manipulation, testing and evidence collection.
Member 4	Security Remediation	Analyze findings, apply secure fixes, improve validation / access control and support retesting.
Member 5	UI/QA & Documentation	Improve website design, perform functional QA, track bugs, screenshots and final report.

6. Final Deliverables

- Functional E-Commerce Web Application
- Penetration Testing Findings & Evidence
- Bug / Vulnerability Report
- Vulnerability Remediation Record
- Improved Website Design
- Final Security Assessment Report & Presentation